Interview Questions and Answers

Network Security | Cyber Security | Cloud Computing

Simple English answers. Each answer has 6 easy points and 1 real example.

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Total questions answered: 109

Tip for the interview: First give a one-line meaning, then say 3 to 4 points in your own words, and finish with the example. Speak slowly and keep your sentences short.

Part 1: Network Security

1. What is a network?

1. A network is a group of two or more devices connected together to share data and resources.
2. The devices can be computers, phones, printers, servers and more.
3. They are connected using cables (wired) or radio waves like Wi-Fi (wireless).
4. The main goal is to share information, files, internet and hardware like printers.
5. Every device in a network has an address so that data reaches the right device.
6. Networks can be small (one room) or very big (the whole world, like the internet).

> **Example:** In a college lab, 30 computers are connected to one printer and one server. Students can print and save files on the server. This lab setup is a network.

2. What is networking?

1. Networking is the process of connecting devices and managing how they talk to each other.
2. It includes designing the network, setting it up, running it and fixing problems.
3. It uses hardware such as routers and switches, and rules called protocols.
4. It makes communication possible, such as email, video calls and file sharing.
5. A person who does this work is called a network engineer or network administrator.
6. Good networking gives faster communication, lower cost and better security.

> **Example:** When an office manager connects all laptops to a Wi-Fi router, sets a password and gives each laptop an IP address, that whole work is called networking.

3. Explain the types of networks.

1. PAN (Personal Area Network): a very small network around one person, within about 10 meters.
2. LAN (Local Area Network): covers a small area like a home, school or office.
3. MAN (Metropolitan Area Network): covers a city or large campus.
4. WAN (Wide Area Network): covers countries or the whole world.
5. Other types are WLAN (wireless LAN), VPN (private network over the internet) and SAN (storage network).
6. We choose the type based on the area to cover, speed needed and cost.

> **Example:** Bluetooth earphones connected to a phone form a PAN. Wi-Fi in a house is a LAN. A cable TV network in a city is a MAN. The internet is a WAN.

4. Explain LAN, MAN, WAN.

1. LAN covers a small area such as a home, lab or one building, usually up to a few kilometers.
2. LAN is fast, cheap, and owned by one person or company.
3. MAN covers a whole city or town, about 5 to 50 kilometers.
4. MAN is used by cable TV companies, banks and city governments to connect their branches.
5. WAN covers a very large area like many countries, and the internet is the biggest WAN.
6. WAN is slower and costlier than LAN, and uses telephone lines, satellites and fiber cables.

> **Example:** A school computer lab is a LAN. A bank connecting all its branches inside Visakhapatnam is a MAN. A company connecting its offices in India and the USA uses a WAN.

5. What is topology and what are the types of topology?

1. Topology means the way devices are arranged and connected in a network.
2. There are two kinds: physical topology (real layout of cables) and logical topology (how data flows).
3. Bus topology: all devices share one main cable.
4. Star topology: all devices connect to one central device like a switch.
5. Ring topology: each device connects to two neighbors, making a circle.
6. Mesh topology and Tree topology are other common types.

> **Example:** In most offices today, every computer is connected to a central switch with its own cable. This layout is a star topology.

6. What are the differences between Bus and Star topology?

1. Layout: Bus uses one single main cable (backbone). Star uses a central switch or hub with a separate cable to each device.
2. Failure: If the main cable breaks in Bus, the whole network stops. In Star, if one cable breaks, only that device is affected.
3. Cost: Bus needs less cable, so it is cheaper. Star needs more cable and a central device, so it costs more.
4. Speed: Bus becomes slow when many devices are added. Star stays fast because each device has its own line.
5. Troubleshooting: Bus is hard to fix as it is difficult to find the fault. Star is easy because we can check each cable separately.
6. Use today: Bus is old and rarely used. Star is the most common topology now.

> **Example:** A small old lab with 5 computers on one cable is a Bus. A modern office where every desk has a cable going to one switch is a Star.

7. What is Ring topology?

1. In Ring topology, each device is connected to exactly two other devices, forming a closed circle.
2. Data travels in one direction (or both in a dual ring) from device to device.
3. Each device passes the data to the next one until it reaches the right receiver.
4. A special signal called a token is often used so that only one device sends data at a time.
5. Advantage: fewer data collisions and fair access for all devices.
6. Disadvantage: if one device or cable fails, the whole ring can stop (unless it is a dual ring).

> **Example:** Token Ring networks in some old offices worked like this. A token went round the circle, and only the computer holding the token could send data.

8. What is Mesh topology?

1. In Mesh topology, every device is connected to every other device directly.
2. In a full mesh with n devices, the number of links is n(n-1)/2.
3. Advantage: very reliable, because if one link fails, data takes another path.
4. Advantage: no traffic problem, as each link carries data of only its two devices.
5. Disadvantage: very costly and hard to set up because of too many cables and ports.
6. It is used where reliability matters most, like military networks and core internet links.

> **Example:** If 4 computers are fully meshed, we need 6 cables (4 x 3 / 2). If one cable fails, the computers can still talk through the other cables.

9. What is Tree topology?

1. Tree topology looks like a tree. It has a root node at the top and branches going down.
2. It is a mix of Bus and Star topology.
3. Many star networks are joined to one main bus cable (backbone).
4. It is easy to add new branches, so it is good for growing networks.
5. It is easy to find and fix faults because each branch is separate.
6. Disadvantage: if the main backbone cable fails, the whole network is cut into parts.

> **Example:** A university has one main switch in the central building. Each department has its own switch connected to it. Every department switch connects to its lab computers. This is a tree topology.

10. What are network devices? Explain about network devices.

1. Network devices are the hardware that connects computers and helps data move in a network.
2. NIC: the card that lets a computer connect to a network.
3. Hub: a simple device that sends data to all connected devices.
4. Switch: a smart device that sends data only to the correct device.
5. Router: connects different networks, such as your home network and the internet.
6. Others are repeater (boosts signal), bridge, gateway, modem and access point.

> **Example:** At home, the modem brings the internet, the router shares it, and the access point gives Wi-Fi to your phone and laptop. All of these are network devices.

11. Explain about NIC, HUB, Switch.

1. NIC (Network Interface Card) is the hardware inside a device that connects it to the network. Every NIC has a unique MAC address.
2. A NIC can be wired (Ethernet port) or wireless (Wi-Fi card).
3. A Hub is a basic device with many ports. It copies the data it receives and sends it out of all ports.
4. Because a hub sends data to everyone, it creates extra traffic and is not secure.
5. A Switch learns the MAC address of each device and sends data only to the device that needs it.
6. A switch is faster, safer and better than a hub, so hubs are almost never used today.

> **Example:** Your laptop's Wi-Fi card is the NIC. When you send a file to your friend through a switch, only your friend's computer gets it. With a hub, all computers would receive it.

12. Explain about router and repeater.

1. A router connects two or more different networks, such as a home network and the internet.
2. A router reads the IP address in each packet and chooses the best path to send it.
3. A router also gives private IP addresses to devices using DHCP and can act as a basic firewall.
4. A repeater is a device that receives a weak signal, cleans it and sends it again with more power.
5. A repeater increases the distance a signal can travel but does not choose paths.
6. A router works at the Network layer, while a repeater works at the Physical layer.

> **Example:** If Wi-Fi in your bedroom is weak, a Wi-Fi repeater placed in the hallway boosts the signal. The router in the living room sends your data out to the internet.

13. What are the differences between HUB and Switch?

1. Data sending: A hub sends data to all ports. A switch sends data only to the port of the correct device.
2. Intelligence: A hub is a dumb device. A switch is smart and keeps a MAC address table.
3. Speed: A hub shares bandwidth between all devices, so it is slow. A switch gives full bandwidth to each port.
4. Collisions: A hub has many data collisions. A switch avoids collisions.
5. Security: A hub is less secure because everyone sees the data. A switch is more secure.
6. Layer and cost: A hub works at the Physical layer and is cheap. A switch works at the Data Link layer and costs more.

> **Example:** If computer A sends a message to computer B, a hub sends it to B, C and D. A switch sends it only to B.

14. What is Internet, Intranet and Extranet?

1. Internet is a global public network that connects millions of networks and anyone can use it.
2. Intranet is a private network inside one organization, used only by its own staff.
3. Extranet is a private network that an organization shares with selected outsiders, like partners or customers.
4. Internet is open to all, Intranet is closed, and Extranet is partly open with a login.
5. Intranet and Extranet are protected with passwords, firewalls and VPNs.
6. All three use the same technology, such as TCP/IP and web pages.

> **Example:** Browsing Google is Internet. A company's HR portal that only employees can open is an Intranet. A portal where a supplier logs in to see its orders is an Extranet.

15. What is an IP Address and what are the versions?

1. IP address means Internet Protocol address. It is a unique number given to every device on a network.
2. It works like a home address, so data knows where to go.
3. An IP address has two parts: the network part and the host (device) part.
4. It can be static (fixed) or dynamic (changes, given by DHCP).
5. There are two versions: IPv4 and IPv6.
6. IPv4 uses 32 bits and IPv6 uses 128 bits.

> **Example:** An IPv4 address looks like 192.168.1.10. An IPv6 address looks like 2001:db8::1.

16. What is IPv4?

1. IPv4 is the fourth version of Internet Protocol and is the most used version.
2. It is a 32-bit address, written as four numbers separated by dots.
3. Each number is from 0 to 255, for example 172.16.5.20.
4. It can make about 4.3 billion unique addresses.
5. IPv4 addresses are divided into classes A, B, C, D and E.
6. Because the addresses are running out, techniques like NAT and private IPs are used.

> **Example:** 8.8.8.8 is an IPv4 address of Google DNS. It has four parts, and each part is between 0 and 255.

17. What is IPv6?

1. IPv6 is the newer version of Internet Protocol, made to replace IPv4.
2. It is a 128-bit address, written in eight groups of hexadecimal numbers separated by colons.
3. It can make a huge number of addresses (about 340 undecillion), so we will not run out.
4. It has built-in security support through IPsec.
5. It gives faster routing because the header is simpler and there is no need for NAT.
6. It supports auto-configuration, so devices can set their own address.

> **Example:** An IPv6 address looks like 2001:0db8:85a3:0000:0000:8a2e:0370:7334. Every phone, bulb and watch can have its own public address.

18. What are the differences between IPv4 and IPv6?

1. Size: IPv4 is 32 bits. IPv6 is 128 bits.
2. Format: IPv4 uses decimal numbers with dots. IPv6 uses hexadecimal numbers with colons.
3. Address count: IPv4 has about 4.3 billion addresses. IPv6 has almost unlimited addresses.
4. Security: IPv4 needs IPsec added separately. IPv6 has IPsec built in.
5. NAT: IPv4 often needs NAT because of address shortage. IPv6 does not need NAT.
6. Setup: IPv4 uses manual setup or DHCP. IPv6 can auto-configure itself.

> **Example:** 192.168.0.1 is IPv4. fe80::1 is IPv6.

19. Explain about Public IP and Private IP.

1. A Public IP is a unique address on the internet, given by your Internet Service Provider (ISP).
2. A Private IP is used inside a home or office network and cannot be reached directly from the internet.
3. Private IP ranges are 10.0.0.0 to 10.255.255.255, 172.16.0.0 to 172.31.255.255, and 192.168.0.0 to 192.168.255.255.
4. Many devices can use the same private IP in different networks, but a public IP must be unique.
5. A router uses NAT to change private IPs to one public IP when going to the internet.
6. Private IPs are free and safer, while public IPs may cost money.

> **Example:** Your home has one public IP given by the ISP. Your phone gets 192.168.1.5 and your laptop gets 192.168.1.6. These are private IPs.

20. What is WWW?

1. WWW means World Wide Web. It is a collection of web pages connected by links and stored on web servers.
2. It is a service that runs on the internet, but it is not the same as the internet.
3. Web pages are opened using a browser such as Chrome or Firefox.
4. Each page has a unique address called a URL.
5. It uses HTTP or HTTPS to send pages from the server to the browser.
6. Web pages are written in HTML and can include text, images, videos and links.

> **Example:** When you type www.wikipedia.org in a browser, a web page is fetched from a server and shown to you. This is the WWW at work.

21. What is a protocol? What are the types of protocols?

1. A protocol is a set of rules that devices follow to communicate with each other.
2. Without protocols, two devices cannot understand each other.
3. Protocols decide data format, speed, error checking and how to start and end a conversation.
4. Web protocols: HTTP and HTTPS.
5. Transfer and mail protocols: FTP, SMTP, POP3, IMAP.
6. Network protocols: TCP, UDP, IP, ARP, ICMP, DNS and DHCP.

> **Example:** Sending an email uses SMTP, receiving it uses IMAP, and opening a website uses HTTP. Each task follows its own protocol.

22. What is HTTP?

1. HTTP means HyperText Transfer Protocol.
2. It is the protocol used to send web pages from a web server to a browser.
3. It works on a request and response model: the browser asks, the server answers.
4. It works at the Application layer and uses port 80.
5. It is stateless, which means the server does not remember earlier requests.
6. Data in HTTP is sent as plain text, so it is not secure.

> **Example:** When you open http://example.com, your browser sends a GET request. The server sends back the page, and the browser shows it.

23. What is HTTPS?

1. HTTPS means HyperText Transfer Protocol Secure.
2. It is the secure form of HTTP and uses port 443.
3. It uses SSL/TLS to encrypt the data between the browser and the server.
4. It uses a digital certificate to prove that the website is genuine.
5. It protects against data theft and tampering.
6. Browsers show a padlock icon for HTTPS sites.

> **Example:** When you pay on an online shopping site, the address starts with https://. Your card number is encrypted so a hacker on the same Wi-Fi cannot read it.

24. What are the differences between HTTP and HTTPS?

1. Security: HTTP is not secure. HTTPS is secure.
2. Encryption: HTTP sends plain text. HTTPS encrypts data using SSL/TLS.
3. Port: HTTP uses port 80. HTTPS uses port 443.
4. Certificate: HTTP needs no certificate. HTTPS needs an SSL/TLS certificate.
5. Speed: HTTP is slightly faster because there is no encryption. HTTPS is a little slower but modern systems make it almost the same.
6. Use: HTTP is fine for simple public pages. HTTPS is a must for logins, banking and shopping.

> **Example:** A login page with http:// can let an attacker see your password. A login page with https:// hides it.

25. What is TCP?

1. TCP means Transmission Control Protocol.
2. It is a connection-oriented protocol. It first makes a connection using a 3-way handshake (SYN, SYN-ACK, ACK).
3. It breaks data into segments and numbers them so they can be put back in order.
4. It checks for errors and re-sends lost data, so delivery is reliable.
5. It has flow control and congestion control to avoid overloading the receiver or network.
6. It is slower than UDP but very reliable.

> **Example:** When you download a file or open a bank website, TCP makes sure every part arrives correctly and in order.

26. What is UDP?

1. UDP means User Datagram Protocol.
2. It is a connectionless protocol. It sends data without making a connection first.
3. It does not check whether the data reached or not, and does not re-send lost data.
4. It has a small header, so it is very fast and light.
5. It is good where speed matters more than perfect delivery.
6. It supports broadcast and multicast.

> **Example:** In a live video call or online game, a few lost packets are fine, but delay is not. So these apps use UDP.

27. What are the differences between TCP and UDP?

1. Connection: TCP is connection-oriented. UDP is connectionless.
2. Reliability: TCP is reliable and re-sends lost data. UDP is unreliable and does not re-send.
3. Speed: TCP is slower. UDP is faster.
4. Order: TCP keeps data in order. UDP does not guarantee order.
5. Header size: TCP header is 20 bytes or more. UDP header is only 8 bytes.
6. Uses: TCP is used for web, email and file transfer. UDP is used for video streaming, VoIP, gaming and DNS.

> **Example:** Sending an email uses TCP so nothing is lost. Watching a live cricket match online uses UDP for speed.

28. Explain about remote login in extranet.

1. Remote login means a person logs in to a private network from a faraway place using the internet.
2. In an extranet, outside users like partners or vendors get limited access to some company resources.
3. They must prove who they are using username, password, and often a one-time code (MFA).
4. A VPN is used to create a safe encrypted tunnel between the user and the company.
5. Firewalls and access rules decide what that outside user can and cannot see.
6. All activity is logged so the company can check who did what.

> **Example:** A supplier logs in to the company's extranet through a VPN, with a password and an OTP. He can only see his own purchase orders and not other company data.

29. What is the OSI reference model? What are the layers in the OSI model?

1. OSI means Open Systems Interconnection. It was made by ISO as a standard guide for network communication.
2. It divides network communication into 7 layers, and each layer has its own job.
3. The layers from top to bottom are: Application, Presentation, Session, Transport, Network, Data Link, Physical.
4. A trick to remember: "All People Seem To Need Data Processing".
5. When sending, data moves down the layers. When receiving, it moves up.
6. It helps in learning, designing and troubleshooting networks, and lets products of different companies work together.

> **Example:** When you send a WhatsApp message, the text is prepared at the top layers, broken into pieces in the middle layers, and sent as signals at the Physical layer.

30. What is the Application layer? Explain the services of the Application layer.

1. The Application layer is Layer 7, the top layer of the OSI model.
2. It is the layer closest to the user. It gives network services to applications like browsers and email apps.
3. Service: Web access using HTTP/HTTPS.
4. Service: Email using SMTP, POP3 and IMAP.
5. Service: File transfer using FTP and remote login using Telnet or SSH.
6. Service: Name lookup using DNS and automatic IP setup using DHCP.

> **Example:** When you open Chrome and type a website name, the browser uses the Application layer to request the page.

31. Explain about protocols in the Application layer.

1. HTTP/HTTPS: used to open web pages.
2. FTP (File Transfer Protocol): used to upload and download files between computers.
3. SMTP: used to send email. POP3 and IMAP are used to receive email.
4. DNS (Domain Name System): changes website names into IP addresses.
5. DHCP: gives IP addresses to devices automatically.
6. Telnet and SSH: used to log in remotely to another computer. SSH is the secure one.

> **Example:** When you type www.google.com, DNS finds its IP address, and then HTTPS fetches the page.

32. What is the Presentation layer? Explain the functionality of the Presentation layer.

1. The Presentation layer is Layer 6 of the OSI model.
2. It prepares data so the Application layer of the receiver can understand it.
3. Function 1: Data translation (changing the format, such as ASCII to Unicode).
4. Function 2: Data compression (making data smaller).
5. Function 3: Data encryption and decryption (keeping data secret).
6. It is sometimes called the "translator" of the network.

> **Example:** If one computer sends text in one coding format and another uses a different one, the Presentation layer converts it so the text looks correct.

33. Explain about data translation in the Presentation layer.

1. Data translation means changing data from one format to another so both sides understand it.
2. Different computers may store data in different ways, for example ASCII, EBCDIC or Unicode.
3. The sender's Presentation layer converts data into a common format for the network.
4. The receiver's Presentation layer converts it back into its own format.
5. It also handles file formats such as JPEG, PNG, MP3 and MPEG.
6. Without it, text could look like strange symbols.

> **Example:** An old mainframe uses EBCDIC and a PC uses ASCII. The Presentation layer converts between them so the message looks the same on both.

34. Explain about data compression in the Presentation layer.

1. Data compression means reducing the size of data before sending it.
2. Smaller data uses less bandwidth and is sent faster.
3. It also saves storage space.
4. Lossless compression keeps all data, as in ZIP and PNG.
5. Lossy compression removes some less important data, as in JPEG and MP3.
6. The receiver's Presentation layer decompresses the data to get it back.

> **Example:** A 10 MB photo is compressed to 2 MB as a JPEG before sending on WhatsApp, so it uploads faster.

35. Explain about data encryption in the Presentation layer.

1. Encryption changes readable data (plain text) into unreadable data (cipher text).
2. Its goal is to keep data safe from hackers while it travels.
3. The sender encrypts the data with a key.
4. The receiver decrypts it with the correct key to read it again.
5. SSL/TLS is the common method that does this for web traffic.
6. It gives confidentiality, and with other tools can also protect integrity.

> **Example:** When you enter your ATM PIN in a banking app, the app encrypts it. Even if someone catches it, they only see random characters.

36. Explain about protocols in the Presentation layer.

1. SSL/TLS: encrypts data for secure communication.
2. JPEG, PNG, GIF: image formats handled at this layer.
3. MPEG, MP3, MIDI: video and audio formats.
4. ASCII, EBCDIC, Unicode: character coding systems for text.
5. XDR and MIME: help in data representation and email attachments.
6. These formats and protocols make sure data is shown correctly on the receiver's device.

> **Example:** When you view a .jpg photo sent by a friend, the JPEG format at this layer helps your phone to show the image properly.

37. What is the Session layer? Explain the functionality of the Session layer.

1. The Session layer is Layer 5 of the OSI model.
2. It starts, manages and ends the sessions (conversations) between two devices.
3. Function 1: Session establishment, which means setting up the connection.
4. Function 2: Session maintenance, which means keeping the connection active.
5. Function 3: Synchronization, which means adding checkpoints, so if a transfer fails it restarts from the last checkpoint.
6. Function 4: Session termination, which means closing the connection properly. It also handles dialog control (half-duplex or full-duplex).

> **Example:** While downloading a big file, a checkpoint is saved every few MB. If the internet drops, the download resumes from the last checkpoint and not from the beginning.

38. What are the differences between authentication and authorization?

1. Meaning: Authentication checks who you are. Authorization checks what you are allowed to do.
2. Order: Authentication comes first. Authorization comes after it.
3. Method: Authentication uses passwords, OTP, fingerprint or face. Authorization uses roles, permissions and access rules.
4. Question: Authentication asks "Who are you?". Authorization asks "What can you access?".
5. Visibility: The user gives information in authentication. In authorization, the system decides and the user does not change it.
6. Example of failure: Wrong password stops you at authentication. Access denied to a file is an authorization failure.

> **Example:** You log in to your office computer with a password (authentication). You can open your own files but not the manager's salary file (authorization).

39. What is an API? Explain about NetBIOS API.

1. API means Application Programming Interface. It is a set of rules that lets one software talk to another software.
2. It hides the inside details and shows only what the programmer needs.
3. It saves time because developers reuse existing functions.
4. NetBIOS means Network Basic Input/Output System.
5. NetBIOS API lets applications on different computers in a LAN communicate, share files and printers.
6. It gives name service, session service and datagram service. It works at the Session layer.

> **Example:** A weather app uses a weather API to get today's temperature. In an old Windows office LAN, NetBIOS lets computers find each other by name and share folders.

40. What is the Transport layer? Explain the functionality of the Transport layer.

1. The Transport layer is Layer 4 of the OSI model.
2. It gives end-to-end communication between the sender and the receiver.
3. Segmentation: it breaks large data into small segments and joins them at the receiver.
4. Port addressing: it uses port numbers to deliver data to the correct application.
5. Flow control and error control: it makes sure the receiver is not overloaded and lost data is re-sent (in TCP).
6. Connection control: it can be connection-oriented (TCP) or connectionless (UDP).

> **Example:** When you open a website and play music at the same time, the Transport layer uses different port numbers so the web data and music data go to the right apps.

41. What is a Segment?

1. A segment is the data unit used at the Transport layer.
2. When the Transport layer receives big data from the upper layers, it cuts it into smaller parts. Each part with a header is a segment.
3. The header has the source port, destination port, sequence number and checksum.
4. The sequence number helps the receiver put the segments in the right order.
5. In UDP, the data unit is called a datagram instead of a segment.
6. Segments are then passed to the Network layer, where they are put inside packets.

> **Example:** A 1 MB file is split into many segments. Each segment is numbered 1, 2, 3 and so on. The receiver joins them in order to rebuild the file.

42. Explain protocols in the Transport layer.

1. TCP is a reliable, connection-oriented protocol.
2. UDP is a fast, connectionless protocol.
3. SCTP (Stream Control Transmission Protocol) supports sending many streams of data at the same time.
4. DCCP is a protocol for time-sensitive data without guaranteed delivery.
5. TCP uses a 3-way handshake, sequence numbers and acknowledgements.
6. UDP is used for video, voice and DNS, and TCP is used for web, email and file transfer.

> **Example:** Downloading a PDF uses TCP so every byte is correct. A live video call uses UDP so there is no delay.

43. What is the Network layer? Explain the functionality of the Network layer.

1. The Network layer is Layer 3 of the OSI model.
2. It moves data from the source network to the destination network, even across many networks.
3. Logical addressing: it adds IP addresses of sender and receiver.
4. Routing: it finds the best path for the data.
5. Packetizing: it puts segments into packets.
6. Fragmentation and error reporting (ICMP) are also done here. Routers work at this layer.

> **Example:** When you send a message from India to a friend in the USA, routers on the way read the destination IP and keep forwarding the packet until it reaches him.

44. Explain about routing and path determination.

1. Routing means moving packets from one network to another using routers.
2. Path determination means choosing the best path out of many possible paths.
3. A router keeps a routing table that lists networks and the next hop for each.
4. The best path is chosen using things like shortest distance, speed, cost and traffic.
5. Static routing means paths are set manually. Dynamic routing means routers learn paths by themselves using protocols such as RIP, OSPF and BGP.
6. If one path fails, a good routing setup finds another path automatically.

> **Example:** If a link between Mumbai and Delhi is down, the router sends the data through Hyderabad instead. The user does not even notice.

45. Explain about logical addressing.

1. Logical addressing means giving a device an address based on software settings, like an IP address.
2. It works at the Network layer (Layer 3).
3. It can be changed. If a device moves to a new network, it gets a new IP address.
4. It has a network part and a host part, which helps routers find the correct network.
5. It can be IPv4 or IPv6.
6. It is used to deliver data from one network to another network.

> **Example:** When you carry your laptop from home to college, your IP address changes from 192.168.1.5 to 10.10.4.20. That is logical addressing.

46. Explain about physical addressing.

1. Physical addressing means the hardware address of a device, called the MAC address.
2. It works at the Data Link layer (Layer 2).
3. It is 48 bits long, written as 6 pairs of hex digits like 00:1A:2B:3C:4D:5E.
4. It is fixed by the manufacturer in the network card.
5. It is used to deliver data between devices on the same local network.
6. It does not change when you move the device to another network.

> **Example:** Inside your home Wi-Fi, the router uses the MAC address of your phone to send the data to your phone and not to any other device.

47. What are the differences between logical addressing and physical addressing?

1. Meaning: Logical address is the IP address. Physical address is the MAC address.
2. Layer: Logical works at Network layer. Physical works at Data Link layer.
3. Size: IPv4 is 32 bits and IPv6 is 128 bits. MAC is 48 bits.
4. Change: Logical address can change. Physical address is normally fixed.
5. Given by: Logical address is given by the network admin or DHCP. Physical address is given by the manufacturer.
6. Use: Logical address is used to reach other networks. Physical address is used inside the same network.

> **Example:** A courier uses your city and street (like IP) to reach your area, and then uses your house door number (like MAC) to reach you.

48. What is a MAC address?

1. MAC means Media Access Control address.
2. It is a unique 48-bit hardware address of a network card (NIC).
3. It is written as 6 groups of hex numbers, such as 3C:5A:B4:12:9F:7D.
4. The first half shows the manufacturer (OUI) and the second half is a unique number for the device.
5. It works at the Data Link layer and is used to deliver data inside a LAN.
6. It can be faked (MAC spoofing), so it should not be the only security check.

> **Example:** A router with MAC filtering lets only your family's devices connect to Wi-Fi by checking their MAC addresses.

49. What is the Physical layer?

1. The Physical layer is Layer 1, the lowest layer of the OSI model.
2. It deals with the real hardware such as cables, connectors, hubs and Wi-Fi signals.
3. It sends raw bits (0s and 1s) as electrical signals, light or radio waves.
4. It decides the data rate, voltage levels and type of cable.
5. It also defines the topology and the transmission mode (simplex, half-duplex, full-duplex).
6. Devices working here are hubs, repeaters, cables and NICs.

> **Example:** When you plug an Ethernet cable into your laptop, the bits move as electrical signals through the cable. That work is done at the Physical layer.

50. What is an encryption algorithm? What are the types of encryption algorithms?

1. An encryption algorithm is a set of math steps that changes readable data into unreadable data using a key.
2. The readable data is plain text and the unreadable data is cipher text.
3. Decryption is the reverse process, getting the original data back with the right key.
4. Symmetric algorithms use one same key for both. Examples are AES, DES and 3DES.
5. Asymmetric algorithms use two keys, a public key and a private key. Examples are RSA and ECC.
6. Hashing (like SHA-256) is also used, but it is one-way and cannot be reversed.

> **Example:** WhatsApp encrypts your messages so only you and the receiver can read them.

51. What is a symmetric algorithm?

1. A symmetric algorithm uses only one secret key for both encryption and decryption.
2. The sender and receiver must both have the same key.
3. It is very fast, so it is good for encrypting large amounts of data.
4. The main problem is how to share the key safely.
5. If the key is stolen, anyone can read the data.
6. Common examples are AES, DES, 3DES and Blowfish.

> **Example:** You lock a box with a key and send the box. Your friend also has a copy of the same key to open it. The same key locks and unlocks. That is symmetric.

52. What is an asymmetric algorithm?

1. An asymmetric algorithm uses two different keys: a public key and a private key.
2. The public key can be shared with anyone and is used to encrypt.
3. The private key is kept secret and is used to decrypt.
4. Key sharing is safe, because the private key never travels.
5. It is slower than symmetric encryption, so it is used for small data like keys.
6. Examples are RSA, ECC and Diffie-Hellman. It is also used for digital signatures.

> **Example:** Rahul gives everyone an open padlock (public key). Anyone can lock a message with it, but only Rahul has the key (private key) to open it.

53. What are the differences between symmetric and asymmetric algorithms?

1. Keys: Symmetric uses one key. Asymmetric uses two keys (public and private).
2. Speed: Symmetric is fast. Asymmetric is slower.
3. Key sharing: Symmetric has a risk when sharing the key. Asymmetric is safer because the public key can be shared openly.
4. Use: Symmetric is used for large data. Asymmetric is used for small data, key exchange and signatures.
5. Key length: Symmetric keys are shorter, like 128 or 256 bits. Asymmetric keys are longer, like 2048 bits.
6. Examples: Symmetric - AES, DES. Asymmetric - RSA, ECC.

> **Example:** HTTPS uses both. It uses asymmetric encryption to safely share a key, and then uses that key for fast symmetric encryption of the web data.

54. Explain about public key and private key.

1. A public key and a private key are a pair of keys made together using math.
2. The public key is shared openly. Anyone can have it.
3. The private key is kept secret and never shared.
4. Data encrypted with the public key can only be opened with the matching private key. This gives confidentiality.
5. Data signed with the private key can be checked with the public key. This proves who sent it.
6. If the private key is lost or stolen, the security is broken, so it must be stored safely.

> **Example:** Your email id is like a public key; everyone knows it and can send you mail. Your password is like the private key; only you can open your inbox.

55. What is a digital signature?

1. A digital signature is an electronic way to prove that a message or document is real and was not changed.
2. It works like a handwritten signature, but it uses math and keys.
3. The sender makes a hash of the message and encrypts it with his private key. This is the signature.
4. The receiver decrypts the signature with the sender's public key and compares the hash.
5. It gives authentication (who sent it), integrity (not changed) and non-repudiation (sender cannot deny).
6. It is used in software updates, e-filing, contracts and SSL certificates.

> **Example:** When you e-sign a PDF contract, the signature shows it came from you. If anyone changes even one letter later, the signature becomes invalid.

56. Explain about firewall in network security.

1. A firewall is a security device or software that watches network traffic and allows or blocks it using rules.
2. It acts like a security guard between a trusted network (inside) and an untrusted network (internet).
3. Packet-filtering firewall checks IP address and port of each packet.
4. Stateful firewall tracks active connections and allows only valid replies.
5. Proxy (application) firewall checks traffic at the application level. Next-generation firewalls (NGFW) also detect threats and apps.
6. Firewalls can be hardware, software or cloud-based.

> **Example:** A company firewall blocks all incoming traffic to port 23 (Telnet) and allows only web traffic on ports 80 and 443. This stops attackers from using the insecure Telnet service.

Part 2: Cyber Security

57. What is cyber security? Explain the advantages of cyber security.

1. Cyber security means protecting computers, networks, apps and data from attacks, damage and theft.
2. It protects three main things: Confidentiality (only allowed people see data), Integrity (data is not changed) and Availability (data is there when needed). This is the CIA triad.
3. Advantage: It protects personal and company data like passwords, bank details and photos.
4. Advantage: It stops hackers, viruses and ransomware from damaging systems.
5. Advantage: It saves money by avoiding data loss, downtime and legal fines.
6. Advantage: It builds trust with customers and helps companies follow laws and rules.

> **Example:** A bank uses firewalls, strong passwords and OTP. Because of this, a hacker cannot easily steal customer money, and customers trust the bank more.

58. Who is a Hacker? Explain the different types of hackers.

1. A hacker is a person who uses computer skills to find and use weaknesses in systems.
2. A hacker may be good or bad, depending on the intention.
3. White hat hacker: a good hacker who finds weaknesses with permission to fix them.
4. Black hat hacker: a bad hacker who attacks to steal data or money.
5. Gray hat hacker: in the middle. He may hack without permission but without bad intention.
6. Other types: Script kiddies (beginners using ready tools), Hacktivists (hack for a cause), Red hat, Blue hat and state-sponsored hackers.

> **Example:** A company hires a white hat hacker to test its website. A black hat hacker, on the other side, breaks into a website to steal credit card numbers.

59. Explain about Black hat hacker, Gray hat hacker and White hat hacker.

1. Black hat hacker breaks into systems illegally for money, revenge or fun, and harms others.
2. Black hat hackers write malware, steal data and sell it on the dark web.
3. White hat hacker works legally, with written permission, to find and fix security holes. Also called an ethical hacker.
4. White hat hackers are hired by companies and follow rules and laws.
5. Gray hat hacker finds weaknesses without permission, but usually does not steal; he may tell the owner or ask for a reward.
6. Gray hat work is still illegal in many places because there was no permission.

> **Example:** A black hat steals customers' data from a shop's website. A white hat is hired by the shop to find the bug. A gray hat finds the bug without asking and emails the owner.

60. What are the differences between black hat and gray hat hackers?

1. Intention: Black hat has bad intention. Gray hat has mixed or no bad intention.
2. Permission: Black hat has no permission. Gray hat also has no permission, but may report the issue.
3. Goal: Black hat wants profit, damage or data theft. Gray hat wants to show skill or push the owner to fix the problem.
4. Harm: Black hat causes serious harm. Gray hat usually causes little or no harm.
5. Legal side: Black hat work is always a crime. Gray hat work can also be a crime, but punishments are often lighter.
6. Behavior after hacking: Black hat hides, sells or misuses data. Gray hat may report it to the owner.

> **Example:** A black hat locks a hospital's files and asks for money. A gray hat finds an open database, tells the hospital and asks for a small reward.

61. Explain about offensive and defensive security.

1. Offensive security means acting like an attacker to find weaknesses before real attackers do.
2. Offensive security includes penetration testing, red teaming and vulnerability research.
3. Defensive security means protecting systems and stopping attacks.
4. Defensive security includes firewalls, antivirus, monitoring, patching and incident response.
5. Offensive team is called Red Team and defensive team is called Blue Team.
6. A company needs both. When they work together, it is called Purple Teaming.

> **Example:** The red team tries to break into a company's server in a test. The blue team watches the alerts, blocks the attack and then fixes the weak point.

62. What is Ethical hacking? Explain the domains in ethical hacking.

1. Ethical hacking means legally testing a system for weaknesses, with the owner's permission, to make it safer.
2. Ethical hackers follow rules, report all findings and do not misuse data.
3. Domain: Network security testing, which checks firewalls, routers and open ports.
4. Domain: Web application security, which checks for SQL injection, XSS and weak login pages.
5. Domain: Wireless, mobile and cloud security testing.
6. Domain: Social engineering tests, malware analysis, digital forensics and IoT security.

> **Example:** A company hires an ethical hacker to test its online shop. He finds that the login page can be bypassed, reports it, and the company fixes it before a real hacker finds it.

63. Explain about Bug bounty hunter, VAPT, SOC.

1. A Bug bounty hunter is a person who finds security bugs in a company's app or website and reports them to get a reward.
2. Companies like Google, Facebook and Microsoft run bug bounty programs on platforms such as HackerOne and Bugcrowd.
3. VAPT means Vulnerability Assessment and Penetration Testing.
4. Vulnerability Assessment finds weaknesses using scanners. Penetration Testing tries to actually use those weaknesses to see the real damage.
5. SOC means Security Operations Center. It is a team and a room that watches the company's systems 24 x 7 for attacks.
6. SOC analysts use tools like SIEM to see alerts, investigate and respond to incidents.

Example: A student finds a bug in a website that exposes user emails, reports it and gets a reward of money. This is bug bounty. A SOC team in a bank sees strange logins at 3 AM and blocks them. This is SOC work.

64. What is an attack? Explain the different types of attacks.

1. An attack is any attempt to harm, steal from or take control of a computer system or network.
2. Attacks target confidentiality, integrity or availability of data.
3. Malware attacks: virus, worm, trojan, ransomware and spyware.
4. Network attacks: DoS and DDoS, Man-in-the-Middle, sniffing and spoofing.
5. Web attacks: SQL injection and cross-site scripting (XSS).
6. Human-based attacks: phishing and social engineering. Password attacks like brute force are also common.

Example: In a DDoS attack, thousands of infected computers send requests to a website at the same time, so the site crashes and real customers cannot use it.

65. Explain about passive attacks and active attacks.

1. A passive attack only watches or listens to data. It does not change anything.
2. Passive attacks are hard to detect because nothing changes in the system.
3. The goal of a passive attack is to steal information, such as passwords or messages.
4. An active attack changes, adds or destroys data or disturbs system services.
5. Active attacks are easier to detect because damage or change is seen.
6. Passive attacks are stopped mostly by encryption. Active attacks are stopped by firewalls, authentication and monitoring.

Example: Someone secretly reading your messages on public Wi-Fi is a passive attack. Someone who changes the amount in your bank transfer is an active attack.

66. Explain about different types of passive attacks.

1. Eavesdropping or sniffing: capturing data as it moves in the network using tools like Wireshark.
2. Traffic analysis: studying who talks to whom, how often and how much, even if data is encrypted.
3. Release of message contents: reading the real content of emails, calls or files.
4. Footprinting or reconnaissance: collecting information about a target before attacking.
5. Passive wiretapping: secretly listening on a cable or wireless link.
6. Defence: use strong encryption like HTTPS and VPN, and avoid open public Wi-Fi.

> **Example:** A hacker sits in a cafe and sniffs the open Wi-Fi to read the websites people visit. He is not changing anything, only listening.

67. Explain about different types of active attacks.

1. Masquerade attack: the attacker pretends to be a real user to get access.
2. Replay attack: the attacker captures valid data and sends it again later.
3. Modification of messages: the attacker changes the content of a message.
4. Denial of Service (DoS/DDoS): the attacker floods a system so real users cannot use it.
5. Man-in-the-Middle (MITM): the attacker sits between two users, reads and changes the data.
6. Others are SQL injection, session hijacking and malware injection.

> **Example:** In a replay attack, a hacker records the signal of your car key and plays it again later to open your car.

68. What are the differences between passive attacks and active attacks?

1. Action: Passive attack only watches. Active attack changes or damages.
2. Data change: Passive makes no change. Active changes data or system.
3. Detection: Passive is hard to detect. Active is easier to detect.
4. Threat to: Passive threatens confidentiality. Active threatens integrity and availability.
5. Prevention: Passive is prevented by encryption. Active is prevented by firewalls, authentication and detection tools.
6. Examples: Passive - sniffing, traffic analysis. Active - DoS, replay, masquerade.

> **Example:** Reading someone's email in secret is passive. Deleting or changing the email is active.

69. What is a cyberattack? Explain about malware.

1. A cyberattack is a bad action done through computers or networks to steal, damage or lock data and systems.
2. It can be done by one hacker, a group or even a country.
3. Common targets are banks, hospitals, companies, governments and normal users.
4. Malware means "malicious software". It is a harmful program made to damage or take control of a system.
5. Malware can steal data, delete files, spy on users, or lock the computer for money.
6. It enters through bad email attachments, fake downloads, infected USB drives and unsafe websites.

> **Example:** A user downloads a free cracked game and, with it, installs malware that secretly steals his saved passwords. This is a cyberattack done through malware.

70. What is malware? Explain about different types of malware.

1. Malware is any software designed to harm a computer, network or user.
2. Virus: attaches to a file and spreads when the file is opened.
3. Worm: spreads by itself through networks without human help.
4. Trojan horse: looks like a useful program but secretly does harm.
5. Ransomware: locks files and asks for money. Spyware: secretly collects user information.
6. Others: adware (shows unwanted ads), rootkit (hides deep in the system), keylogger (records typing) and botnet (group of infected computers).

> **Example:** A keylogger installed on a computer records every key you press, including your bank password, and sends it to the attacker.

71. Explain about spyware.

1. Spyware is a malware that secretly watches user activity and sends the data to an attacker.
2. It can record browsing history, passwords, keystrokes, emails and screen.
3. It usually runs in the background, so the user does not know.
4. It enters through free software, bad links and fake apps.
5. Signs are slow computer, strange pop-ups, battery drain and unknown programs.
6. Protection: use antivirus, install apps from trusted sources only and keep software updated.

> **Example:** A free "PDF converter" app also installs spyware. It takes screenshots of your screen and sends them to a hacker.

72. Explain about Virus.

1. A virus is a malicious program that attaches itself to a clean file or program.
2. It needs human action to spread, like opening an infected file or running a program.
3. It copies itself into other files and can damage or delete data.
4. It spreads through email attachments, USB drives and downloads.
5. Types include file virus, boot sector virus, macro virus and polymorphic virus.
6. Protection: antivirus software, regular updates, and not opening unknown attachments.

> **Example:** A user opens an email attachment named "invoice.doc" that has a macro virus. The virus copies itself into other Word files and corrupts them.

73. Explain about Worm.

1. A worm is a malware that copies itself and spreads automatically through networks.
2. It does not need a host file and does not need any user action.
3. It uses weaknesses in the operating system or software to spread.
4. It can use a lot of bandwidth and slow down or crash networks.
5. It may also carry other harmful code, like opening a back door for hackers.
6. Protection: install security patches, use firewalls and close unused ports.

> **Example:** The WannaCry worm in 2017 spread by itself to hundreds of thousands of Windows computers around the world without anyone clicking on anything.

74. Explain about Trojan horse.

1. A Trojan horse is a malware that pretends to be a useful or harmless program.
2. The name comes from the old Greek story of a wooden horse with soldiers hidden inside.
3. It does not copy itself like a virus or worm. The user installs it himself by mistake.
4. After installing, it can steal data, open a back door, or install other malware.
5. Types include backdoor trojan, banking trojan and remote access trojan (RAT).
6. Protection: download software only from trusted sources and do not use cracked software.

> **Example:** A user downloads a free "video player" that works well, but it secretly gives the hacker remote control of the computer.

75. Explain about ransomware.

1. Ransomware is a malware that locks or encrypts your files and asks for money (ransom) to unlock them.
2. The ransom is usually asked in cryptocurrency like Bitcoin.
3. It enters mostly through phishing emails, fake downloads and weak remote access.
4. Paying does not always give the files back, and it encourages more attacks.
5. Hospitals, schools and companies are common targets because they cannot afford downtime.
6. Protection: take regular offline backups, update systems and train staff about phishing.

> **Example:** In the WannaCry attack, files on computers were locked and a message asked for Bitcoin. Companies that had backups restored their data without paying.

76. What are the differences between virus and worm?

1. Spread: A virus needs a host file and human action. A worm spreads by itself.
2. Host: A virus attaches to a file or program. A worm is a standalone program.
3. Speed: A virus spreads slowly. A worm spreads very fast.
4. Target: A virus mainly harms files and programs. A worm mainly harms networks and uses bandwidth.
5. Medium: A virus spreads through shared files, USB and email attachments. A worm spreads through network weaknesses.
6. Control: A virus is stopped by antivirus. A worm needs patches, firewalls and antivirus.

> **Example:** A virus spreads when you open an infected Word file. A worm like WannaCry spreads on its own to every computer in the network.

77. What is pretexting? Explain the various types of attacks under pretexting.

1. Pretexting means the attacker makes up a false story or identity to trick a person into giving information.
2. The attacker builds trust first, then asks for secret data.
3. Phishing: fake emails or messages that pretend to be from trusted sources.
4. Vishing: fake phone calls that pretend to be from a bank or company.
5. Smishing: fake SMS messages with bad links.
6. Impersonation or baiting: pretending to be an IT person, police officer or delivery agent, or leaving infected USB drives for people to pick up.

> **Example:** A caller says, "I am from the IT department. Your account has a problem. Please tell me your password to fix it." This is pretexting.

78. What is a phishing attack? Explain with a real-time example.

1. Phishing is a cyberattack where the attacker sends a fake message that looks real to trick you into giving private information.
2. The message may ask for passwords, OTP, card numbers or ask you to click a bad link.
3. Attackers use urgent words like "Your account will be blocked today".
4. The link opens a fake website that looks exactly like the real one.
5. Phishing is the most common way for attackers to enter a company.
6. Protection: check the sender address, do not click unknown links, and use two-factor authentication.

> **Example:** You get an SMS: "Your SBI account is blocked. Click here to update KYC." The link opens a copy of the bank site. When you enter your login and OTP, the attacker uses them to steal your money.

79. Explain about different types of phishing attacks.

1. Email phishing: mass fake emails sent to many people.
2. Spear phishing: a targeted attack on one person or a small group, using their personal details.
3. Whaling: a spear phishing attack aimed at top bosses like the CEO or CFO.
4. Vishing (voice phishing) and Smishing (SMS phishing).
5. Clone phishing: the attacker copies a real email you received and replaces the link with a bad one.
6. Pharming and angler phishing (through fake social media support accounts) are also types.

> **Example:** An attacker sends an email to the finance manager that looks like it is from the CEO: "Please send Rs. 5 lakh to this account urgently." This is whaling.

80. Explain about spoofing.

1. Spoofing means pretending to be a trusted person, device or website by faking identity details.
2. The attacker hides his real identity to gain trust or access.
3. Email spoofing: faking the sender's email address.
4. IP spoofing: faking the source IP address of packets.
5. MAC spoofing, DNS spoofing and caller ID spoofing are other types.
6. Protection: use email authentication (SPF, DKIM), firewalls and always verify the sender.

> **Example:** You get an email that shows "support@yourbank.com" as the sender, but it was really sent by a hacker who faked that address. This is email spoofing.

81. Explain about social engineering.

1. Social engineering is tricking people, not computers, to give secret information or access.
2. It uses human feelings like trust, fear, greed, curiosity and urgency.
3. It is called "hacking the human" because people are often the weakest link.
4. Techniques: phishing, pretexting, baiting (leaving infected USB), tailgating (following an employee into a secure door) and quid pro quo (offering a favor for information).
5. It works because it needs no technical skill.
6. Protection: train staff, verify requests through another channel and never share passwords.

> **Example:** An attacker wears a delivery uniform, carries a box and asks a worker to hold the office door open. This is tailgating, and he can now enter the secure area.

82. Explain about pharming.

1. Pharming is an attack that sends users to a fake website even when they type the correct website address.
2. The attacker changes DNS information or the user's host file.
3. Because the user did not click any bad link, it is harder to notice.
4. The fake site looks exactly like the real one and steals login details.
5. It is dangerous for banking and shopping sites.
6. Protection: use HTTPS and check the padlock, keep antivirus updated, use secure DNS and check the web address carefully.

> **Example:** You type your bank's real address in your browser, but because your DNS was poisoned, you land on a fake bank page. You enter your password and the hacker gets it.

Part 3: Cloud Computing

83. What is cloud computing? Explain the advantages of cloud computing.

1. Cloud computing means using servers, storage, software and databases over the internet instead of your own computer or data room.
2. You pay only for what you use, like an electricity bill.
3. Advantage: Low cost, because you do not buy expensive hardware.
4. Advantage: Scalability, so you can increase or decrease resources in minutes.
5. Advantage: Access from anywhere, at any time, with internet.
6. Advantage: Automatic backup, updates and strong security from the provider.

Example: A startup runs its website on AWS. On a sale day, traffic goes up 10 times, so it adds more servers for a few hours and pays only for those hours.

84. Who is a cloud engineer?

1. A cloud engineer is an IT professional who designs, builds and manages systems on the cloud.
2. He works with cloud platforms like AWS, Azure and Google Cloud.
3. He sets up servers, storage, networks and databases in the cloud.
4. He takes care of security, backups, monitoring and cost control.
5. He moves company applications from old data centers to the cloud (cloud migration).
6. Skills needed: Linux, networking, scripting (Python, Bash), DevOps tools and one cloud platform.

Example: A cloud engineer in a bank creates virtual servers on AWS, sets firewall rules, and makes sure the system keeps running even if one server fails.

85. What is cloud? List some different versions of cloud.

1. Cloud is a network of remote servers on the internet that store data and run applications for users.
2. You do not need to know where the physical machine is placed.
3. Public cloud: shared and owned by a provider, such as AWS or Azure.
4. Private cloud: used by only one company.
5. Hybrid cloud: a mix of public and private cloud.
6. Community cloud: shared by a few organizations with the same needs, and Multi-cloud: using more than one provider.

Example: Google Drive and Gmail are on a public cloud. A bank keeping customer data in its own secure cloud uses a private cloud.

86. Explain about different types of service models.

1. Cloud service models show how much work the provider does and how much the user does.
2. IaaS (Infrastructure as a Service): gives virtual servers, storage and network.
3. PaaS (Platform as a Service): gives a ready platform to build and run apps.
4. SaaS (Software as a Service): gives ready-to-use software through a browser.
5. FaaS (Function as a Service): lets you run small pieces of code without managing servers.
6. The higher you go from IaaS to SaaS, the less you manage and the easier it becomes.

Example: Renting a plain empty flat is IaaS. A furnished flat is PaaS. Staying in a hotel is SaaS.

87. Explain briefly about IaaS, PaaS, SaaS.

1. IaaS gives the basic building blocks: virtual machines, storage and networks. The user installs the OS and software.
2. Examples of IaaS: AWS EC2, Azure Virtual Machines, Google Compute Engine.
3. PaaS gives a platform with OS, runtime and tools. The user only writes and deploys code.
4. Examples of PaaS: AWS Elastic Beanstalk, Google App Engine, Azure App Service.
5. SaaS gives complete software over the internet. The user only uses it.
6. Examples of SaaS: Gmail, Microsoft 365, Zoom, Salesforce.

> **Example:** A company that rents a server on AWS and installs its own software uses IaaS. A developer who uploads code to Google App Engine uses PaaS. An employee using Gmail uses SaaS.

88. Explain the benefits of IaaS, PaaS, SaaS.

1. IaaS benefit: full control and flexibility, and no need to buy hardware.
2. IaaS benefit: pay-as-you-go and easy to scale up or down.
3. PaaS benefit: developers focus only on code, because the provider manages servers and updates.
4. PaaS benefit: faster development and easy teamwork.
5. SaaS benefit: no installation or maintenance, and users get automatic updates.
6. SaaS benefit: access from any device, and low starting cost with a monthly fee.

> **Example:** A small business uses SaaS accounting software. It does not need an IT team to install or update anything.

89. What is meant by Function as a Service (FaaS)?

1. FaaS is a cloud model where you write small functions and the cloud runs them when needed.
2. It is also called serverless computing, though servers still exist; the provider manages them.
3. Functions run only when triggered by an event, like a file upload or a button click.
4. You pay only for the time the function runs, even in milliseconds.
5. It scales automatically from one request to thousands.
6. Examples: AWS Lambda, Azure Functions, Google Cloud Functions.

> **Example:** When a user uploads a photo, an AWS Lambda function runs, makes a small thumbnail and then stops. You pay only for those few seconds.

90. Define the various cloud storage levels in cloud computing.

1. Cloud storage is offered in different types based on how data is stored and how often it is used.
2. Object storage: stores data as objects with a key, good for photos, videos and backups. Example: Amazon S3.
3. Block storage: stores data in fixed blocks like a hard disk, good for databases and servers. Example: Amazon EBS.
4. File storage: stores data in files and folders that many servers can share. Example: Amazon EFS.
5. Storage tiers by usage: Hot (frequently used), Cool (sometimes used) and Archive (rarely used, cheapest).
6. Choosing the right level saves cost and gives the right speed.

> **Example:** A company keeps this month's photos in hot storage for fast access, and moves 5-year-old photos to archive storage to save money.

91. What are some of the key features of cloud computing?

1. On-demand self-service: users can get resources by themselves, any time.
2. Broad network access: services are available over the internet from any device.
3. Resource pooling: many users share the same physical resources safely.
4. Rapid elasticity: resources can grow or shrink quickly as needed.
5. Measured service: usage is tracked and you pay for what you use.
6. Reliability and security: data is backed up and often copied to many places.

> **Example:** A video app creates 50 extra servers during a live match and removes them after the match. This is elasticity and pay-per-use.

92. Explain briefly about public, private and hybrid cloud.

1. Public cloud is owned by a provider and shared by many customers over the internet.
2. Public cloud examples: AWS, Azure and Google Cloud. It is cheap and easy to scale.
3. Private cloud is used by only one organization, in its own data center or hosted privately.
4. Private cloud gives more control, security and privacy, but costs more.
5. Hybrid cloud is a mix of both, with data and apps moving between them.
6. Hybrid lets a company keep sensitive data private and use the public cloud for other work.

> **Example:** A hospital keeps patient records in a private cloud and runs its public website on a public cloud. This is hybrid.

93. What are the benefits and limitations of a Private Cloud and a Public Cloud?

1. Private cloud benefit: high security, full control and good for strict rules like banking and health.
2. Private cloud limitation: high cost to build and run, and needs its own IT team.
3. Private cloud limitation: scaling is slower because you own limited hardware.
4. Public cloud benefit: low cost, pay-as-you-go and almost unlimited scaling.
5. Public cloud benefit: no maintenance, since the provider takes care of hardware.
6. Public cloud limitation: less control, shared environment and possible data privacy and compliance worries.

> **Example:** A startup picks public cloud because it has little money. A government defense department picks private cloud for secrecy.

94. What are the benefits of a Hybrid Cloud?

1. Security: keep sensitive data in the private cloud and other data in the public cloud.
2. Flexibility: move workloads between private and public as needs change.
3. Cost saving: use cheap public cloud for normal work and avoid buying extra hardware.
4. Scalability: use "cloud bursting" to take extra power from the public cloud during busy times.
5. Business continuity: use the public cloud as a backup if the private cloud fails.
6. Easy compliance: keep regulated data where laws require, and still use modern cloud tools.

> **Example:** An online shop keeps customer payment data on its private cloud, but during a festival sale it adds public cloud servers to handle huge traffic.

95. Who are the Cloud service providers in a cloud ecosystem?

1. A cloud service provider (CSP) is a company that offers cloud services like servers, storage and software.
2. Amazon Web Services (AWS) is the biggest provider.
3. Microsoft Azure is second in size and is strong in business and Windows.
4. Google Cloud Platform (GCP) is strong in data, AI and containers.
5. Others: IBM Cloud, Oracle Cloud, Alibaba Cloud and Salesforce.
6. Some also offer SaaS products, like Dropbox, Zoom and Slack.

> **Example:** A company can host its website on AWS, use Microsoft 365 for office work and use Google Cloud for data analysis.

96. What is AWS and why is it so popular?

1. AWS means Amazon Web Services. It is the cloud platform of Amazon, started in 2006.
2. It offers over 200 services such as computing, storage, databases, networking, AI and security.
3. It works on a pay-as-you-go model.
4. It has data centers in many regions around the world.
5. It was the first big cloud provider, so it has a large customer base and a big community.
6. Famous customers include Netflix, Airbnb, and many startups and governments.

> **Example:** Netflix uses AWS to stream movies to millions of users around the world without building its own data centers.

97. Explain reasons for why AWS is so popular.

1. First mover: AWS started early and gained trust.
2. Wide range of services: it has tools for almost every need.
3. Global reach: many regions and availability zones near customers.
4. Reliability: high uptime and backup across zones.
5. Flexible pricing: pay-as-you-go, reserved and free tier for learners.
6. Strong security, large community, good documentation and many job opportunities.

> **Example:** A student can use the AWS Free Tier to host a small website for a year at almost no cost, which makes learning easy.

98. What are the AWS services?

1. Compute: EC2 (virtual servers), Lambda (serverless), Elastic Beanstalk.
2. Storage: S3 (object storage), EBS (block storage), Glacier (archive).
3. Database: RDS (relational), DynamoDB (NoSQL), Redshift (data warehouse).
4. Networking: VPC (private network), Route 53 (DNS), CloudFront (CDN).
5. Security: IAM (users and permissions), KMS (keys), Shield (DDoS protection).
6. Management and others: CloudWatch (monitoring), CloudFormation, and AI services like SageMaker.

> **Example:** To host a website, you can run it on EC2, store images in S3, use RDS for the database, and use CloudFront to make it load fast.

99. What is Microsoft Azure Cloud? Microsoft Azure Cloud is famous for which service?

1. Azure is the cloud platform of Microsoft, launched in 2010.
2. It offers computing, storage, databases, networking, AI and analytics.
3. It supports Windows, Linux, and many languages and tools.
4. It is very strong in hybrid cloud, so companies can connect on-premises servers with the cloud.
5. It is famous for Azure Active Directory (now Microsoft Entra ID), Azure Virtual Machines and its link with Microsoft 365.
6. It is popular with large companies that already use Microsoft products.

> **Example:** A company that already uses Windows Server and Office 365 finds it easy to move to Azure, because everything works well together.

100. What are the main services offered by Azure?

1. Compute: Azure Virtual Machines, Azure App Service, Azure Functions and Kubernetes Service (AKS).
2. Storage: Blob Storage, File Storage, Disk Storage.
3. Database: Azure SQL Database, Cosmos DB.
4. Networking: Virtual Network, Load Balancer, Azure DNS and VPN Gateway.
5. Security and identity: Microsoft Entra ID (Azure AD), Key Vault, Defender for Cloud.
6. AI and others: Azure AI services, DevOps, Azure Monitor and Backup.

> **Example:** A company stores its files in Azure Blob Storage, runs its app in App Service, and uses Azure AD so employees log in once to everything.

101. What is a data centre?

1. A data centre is a building or room that holds many servers, storage devices and network equipment.
2. It stores, processes and shares large amounts of data and applications.
3. It needs power backup (UPS and generators), cooling systems and fire protection.
4. It has strong security such as guards, cameras and biometric locks.
5. Network connections are duplicated so service continues even if one link fails.
6. Cloud providers build many data centers around the world and group them into regions and zones.

> **Example:** When you open Gmail, your request goes to a Google data centre, where servers find and send back your emails.

102. What is Google Cloud Platform (GCP)?

1. GCP is the cloud platform of Google, built on the same systems that run Google Search and YouTube.
2. It offers computing, storage, databases, networking, AI and analytics.
3. Main services: Compute Engine, Cloud Storage, BigQuery, Cloud Functions and Google Kubernetes Engine (GKE).
4. It is famous for big data analytics and machine learning.
5. It supports open source tools very well.
6. It has data centers in many regions and uses Google's fast private network.

> **Example:** A company uses BigQuery to study billions of rows of sales data within seconds, and gets reports for its managers.

103. What are the advantages of GCP?

1. Strong data analytics and machine learning tools such as BigQuery and Vertex AI.
2. Very fast and secure global network owned by Google.
3. Kubernetes started at Google, so GKE is one of the best container services.
4. Pricing is simple, with per-second billing and automatic discounts for long use.
5. Good security with encryption by default for data.
6. Friendly to open source and multi-cloud use, and it has a free tier for learning.

Example: A startup working on AI chooses GCP because it has ready-made machine learning tools and a fast network.

104. Explain the main differences between AWS, Azure and GCP.

1. Market share: AWS is the biggest, Azure is second and GCP is third.
2. Strength: AWS has the widest services. Azure is best for Microsoft users and hybrid cloud. GCP is best for data, AI and Kubernetes.
3. Compute service: AWS has EC2, Azure has Virtual Machines, GCP has Compute Engine.
4. Storage: AWS has S3, Azure has Blob Storage, GCP has Cloud Storage.
5. Pricing: AWS has many pricing options. Azure gives benefits to existing Microsoft licence holders. GCP gives automatic discounts and per-second billing.
6. Launch: AWS in 2006, Azure in 2010 and GCP in 2008 (App Engine) with wider services later.

Example: A company using lots of Windows and Office may choose Azure. A data science team may choose GCP. A company needing the widest set of tools may choose AWS.

105. Compare AWS EC2, Azure VM and Google Compute Engine.

1. All three give virtual machines in the cloud, where you choose CPU, RAM, storage and operating system.
2. AWS EC2 is from Amazon. It has many instance types and options like Spot and Reserved instances.
3. Azure VM is from Microsoft. It works well with Windows Server, Active Directory and hybrid setups.
4. Google Compute Engine is from Google. It has custom machine sizes and per-second billing.
5. Scaling: EC2 uses Auto Scaling, Azure uses VM Scale Sets, and GCP uses Managed Instance Groups.
6. Cheap options: EC2 has Spot instances, Azure has Spot VMs, and GCP has Preemptible or Spot VMs.

Example: To run a Linux web server, you can pick any of the three. A company with many Windows apps often picks Azure VM for easy integration.

106. What is virtualization?

1. Virtualization means creating virtual (software-based) versions of hardware like servers, storage or networks.
2. It lets one physical computer run many virtual machines (VMs) at the same time.
3. Each VM has its own operating system and works like a separate computer.
4. It uses hardware better and saves money, space and power.
5. It makes backup, testing and moving of systems easier.
6. It is the base technology of cloud computing.

> **Example:** On one laptop you can use VirtualBox to run Windows and Linux at the same time, as two virtual machines.

107. What is a hypervisor?

1. A hypervisor is software that creates and manages virtual machines.
2. It sits between the hardware and the virtual machines.
3. It shares CPU, memory and storage among the VMs.
4. It keeps the VMs separate from each other, so one VM cannot harm another.
5. It is also called a Virtual Machine Monitor (VMM).
6. Examples: VMware ESXi, Microsoft Hyper-V, KVM, Oracle VirtualBox.

> **Example:** A hypervisor on a server splits it into five VMs. Each VM runs a different app, and they do not disturb each other.

108. What are the two types of hypervisors?

1. Type 1 hypervisor is also called bare-metal hypervisor.
2. Type 1 runs directly on the physical hardware, with no operating system below it.
3. Examples of Type 1: VMware ESXi, Microsoft Hyper-V, Xen and KVM.
4. Type 2 hypervisor is also called hosted hypervisor.
5. Type 2 runs as a normal program on top of an operating system like Windows or Linux.
6. Examples of Type 2: Oracle VirtualBox, VMware Workstation and Parallels.

> **Example:** A cloud company uses Type 1 on its servers for high performance. A student uses Type 2 (VirtualBox) on a laptop to practice Linux.

109. Explain the difference between Type-I and Type-II hypervisors.

1. Installation: Type I is installed directly on hardware. Type II is installed on top of an OS.
2. Speed: Type I is faster as there is no extra OS layer. Type II is slower.
3. Security and stability: Type I is more secure and stable. Type II depends on the host OS, so it is less secure.
4. Use: Type I is used in data centers and clouds. Type II is used for personal use, testing and learning.
5. Cost and setup: Type I needs special setup and skill. Type II is easy to install and use.
6. Examples: Type I - VMware ESXi, Hyper-V. Type II - VirtualBox, VMware Workstation.

> **Example:** AWS runs customer servers on a Type 1 hypervisor. A student testing software on his laptop uses a Type 2 hypervisor.